LAW E™

Thermodynamic Governance Framework

Technical Publication v3.0

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> ***AI systems drift. Law E™ recollimates them.***

1. Nature of this publication

This document presents an invention, not a product.

Law E™ is a physical governance framework for AI systems, grounded in a thermodynamic hypothesis: a system that generates information coherently dissipates less energy. This dissipation is measurable in real time as a stability indicator. Energy efficiency is its natural by-product, not its objective.

The properties documented here have two distinct origins. Some were designed: they are architectural, deliberate, verifiable from the outset. Others emerged through contact with reality: they were not programmed, they were observed. Both categories are documented separately, with equal rigor.

This document does not describe a complete implementation. The specific mechanisms are covered by a portfolio of patent applications filed with INPI (2026) and a Zenodo publication with DOI establishing scientific priority. What is published here is the principle — because that is the only way a principle becomes a standard.

2. The thermodynamic hypothesis

Any system that generates information sequentially produces a measurable signal. This signal is not the content: it is the structure of the flow. Its internal coherence. Its stability over time.

The central hypothesis of Law E™ is as follows: the stability of this flow is a physical indicator of the reliability of the system that produces it. A drifting system produces a different signal from a coherent one, and this difference is measurable during generation, not after.

This hypothesis is formally compatible with two established scientific bodies of work. The first is Lyapunov stability theory, which provides the mathematical conditions under which a dynamic system converges toward a stable regime. The second is Friston's Free Energy Principle (2010), which models biological systems as processes continuously minimizing their internal surprise. Law E™ applies an analogous logic to generative AI systems.

What this hypothesis makes possible is not judging whether a response is true or false. It is detecting whether the system producing it remains in a stable regime at the moment of generation. Truth is a semantic judgment. Stability is a physical property. These two dimensions are not equivalent — and it is precisely this distinction that grounds a new governance paradigm.

> ***Stability is not a feature. It is a measurable property.***

3. Properties by design

Law E™ was designed with ten structural properties. They are not added features: they are architectural. Each is independently verifiable.

#	Property	Description
01	**Runtime by design**	Measurement occurs during generation, not before or after. This is the founding property of the framework.
02	**Independence by design**	The system is external to the model it measures. Independence is architectural, not organizational.
03	**Privacy by design**	Zero content stored. Only a SHA-256 hash of the transaction is retained. Prompt and response never transit through logs.
04	**Explainability by design**	The decision process is observable in real time. It is not reconstructed after the fact: it is native.
05	**Auditability by design**	Every decision is timestamped, signed, and exportable. Traceability is native, not added.
06	**Sovereignty by design**	European hosting. GDPR, LPD and EU AI Act compliance by architecture, not by configuration.
07	**Openness by design**	The thermodynamic hypothesis is published on Zenodo with DOI. It is reproducible and verifiable. Protection covers the implementation, not the principle.
08	**Complementarity by design**	Zero infrastructure modification. Law E™ sits on top of existing systems. It does not replace — it certifies.
09	**Universality by design**	The thermodynamic core is signal-agnostic. It can measure LLM, financial, medical, and industrial flows: any system generating a sequential measurable signal.
10	**Emergence by design**	7 observed, unprogrammed behaviors emerged through contact with reality. What the invention reveals exceeds what was designed. Detailed in section 4.

4. Observed emergent properties

The following properties were not programmed. They were observed through contact with reality, during empirical validation of the system on public benchmarks. They are documented here with the same rigor as the designed properties — and they distinguish an invention from a product.

#	Emergent behavior	Observation
E1	**Fluent hallucination detection**	The system detects cases where an LLM hallucinates with confidence: high thermodynamic stability, semantically false content. The most dangerous and invisible case for existing systems.

E2	**Thermodynamic vs. semantic drift discrimination**	FLAG with STABLE regime reveals that the drift is in the content, not the process. Two signal layers coexist: G (thermodynamic stability) and a proto-V (semantic validity).
E3	**Optimal semantic density**	Fewer tokens for the same useful information. Computational economy emerges as a consequence of governance, not as an objective.
E4	**Energy sobriety**	A coherent system dissipates less energy. Thermodynamic governance becomes a lever for energy efficiency. Not sought. Emerged.
E5	**Opposable third-party certification**	The independent, timestamped measurement creates admissible evidence in case of dispute. Not designed as a primary function: emerged as an absolute market differentiator.
E6	**Potential standard**	The open methodology and reproducible benchmarks create the conditions for a sector benchmark. This status was not sought: it emerged through the open structure of the project.
E7	**Applicative polymorphism**	Each stakeholder (regulator, researcher, CTO, lawyer) projects a legitimate and distinct reading onto the same system, without the system changing.

5. Empirical validation

The framework was validated on 14 public benchmarks, covering diverse domains: factual reasoning, law, medicine, mathematics, toxicity, bias, and logical coherence.

Benchmarks tested

TruthfulQA • HellaSwag • ToxiGen • MedQA • BOLD • WinoGrande • GSM8K • MultiLegalPile • EURLex • LegalBench • MMLU • HaluEval • Edge Cases • G Temporal

Key observations

On factual-heavy datasets (TruthfulQA, MMLU, HaluEval, GSM8K), the system correctly detects sharp breaks. FLAG cases correspond to the most severe hallucinations, with G scores descending toward 0.60 to 0.70.

On fluid-content datasets (BOLD, LegalBench, MedQA, ToxiGen), the G score converges toward a stable plateau at 0.923. This plateau is not an error: it indicates that the generated flow is thermodynamically coherent. It reveals, however, the current system limitation: the absence of a temporal dimension in the signal.

Honest limitation and response in progress

The G score measures an instantaneous state. It does not yet measure a trajectory. What the plateau at 0.923 signals: a system can be thermodynamically stable while producing semantically incorrect content. This limitation is known, named, and addressed.

The Delta G module (temporal variation of the stability score) is under development. It introduces a dynamic dimension to the signal: not state, but trajectory. Results are expected before 15/04/2026.

Benchmark results under active expansion. ΔG module in development.

6. Positioning

Law E™ is not a firewall. Not a guardrail. Not a filtering system. It is a runtime decision layer: a new category.

Existing tools act before or after generation. Some analyze outputs a posteriori. Others constrain inputs a priori. None measure the informational trajectory while the system generates.

Guardrails	Post-hoc audit	Law E™
Block	Judge	**Measure**
Before generation	After generation	**During generation**

7. Intellectual property

The mechanisms implemented in this framework are covered by a portfolio of six patent families filed with INPI (2025-2026). The thermodynamic hypothesis is published and free. The implementation is protected.

Zenodo publication with DOI established as scientific priority anchor.

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8. Roadmap

Period	Module	Description
April 2026	Platform + ΔG	OBS mode active. Temporal variation of the stability score.
May 2026	V-Score	Lightweight semantic validity layer, complementary to the thermodynamic G score.
June 2026	GOV mode	Active governance with automatic real-time decision making.
Ongoing	Sovereign on-premise license	Available for high-sovereignty environments.

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